

TEKIRYA BRIGHTON



PERSONAL DETAILS

 **TEKIRYA BRIGHTON**

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 Kasubi Kampala

 Male

 Uganda

TECHNICAL SKILLS

LANGUAGES

- Python
- C
- PHP
- HTML
- Javascript
- R
- SQL
- C++

AI & ML

- Tensorflow.
- Pytorch.
- OpenCV
- Scikit-learn.
- LLMS
- NLP
- Deep Learning.
- Data Mining.
- Data warehousing.
- Computer Vision.
- Machine Learning.

PROFILE

I am a passionate and technically driven AI/ML Engineer with extensive hands-on experience in building intelligent systems that combine artificial intelligence, machine learning, IoT, and networking technologies. My career has been defined by a commitment to solving real-world problems through innovation, and I have successfully delivered impactful projects across healthcare, agriculture, smart environments, and secure digital ecosystems.

I have developed groundbreaking solutions such as **FetalCare**, an AI-powered fetal health analysis device that supports maternal healthcare, and an **AI vein locator** that assists clinicians in identifying veins for medical procedures. In agriculture, I built a **deep learning-based tomato disease detection system** that helps farmers protect crops and improve yields. I also engineered **smart home automation solutions** using Arduino, Raspberry Pi, and sensor integration, creating environments that are both efficient and user-friendly.

My expertise extends to **networking and cybersecurity**, where I have designed and tested modular network scanning tools, set up ethical penetration testing labs, and developed privacy-first communication systems such as encrypted chatrooms and secure file vaults. I also created a **local document analysis tool for the Uganda Communications Commission (UCC)**, enabling efficient classification and management of regulatory documents.

Technically, I am skilled in **Python, TensorFlow, PyTorch, FastAPI, and full-stack web development with HTML, CSS, and JavaScript**. I have a strong foundation in **computer vision, sensor integration, hardware-software interfacing, and secure networking protocols**. My creative side shines through in hacker-themed, neon-green interfaces and gamified security tools that make complex systems engaging and accessible.

I am deeply committed to advancing intelligent, reproducible AI systems that democratize technology and empower users to control their data. My vision is to continue innovating in healthcare, agriculture, smart environments, and cybersecurity, while mastering large language model development and teaching others through hands-on projects.

EDUCATION

YEAR	SCHOOL/INSTITUTION	AWARD
2023 – 2025	ISBAT UNIVERSITY, UGANDA	BSc. ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
2020 – 2022	KISUBI MAPEERA SENIOR SECONDARY SCHOOL, UGANDA	UGANDA ADVANCED CERTIFICATE OF EDUCATION
2017 – 2019	KIKAAYA COLLEGE SCHOOL, UGANDA	UGANDA CERTIFICATE OF EDUCATION
2014 – 2015	BUDDO PARENTS ACADEMY	PRIMARY LEAVING EXAMINATION

IOT AND EMBEDDED SYSTEMS

- Arduino
- Raspberry Pi
- AVR microchip Studio
- Sensors (PIR, Ultrasonic, Temp)

WEB DEVELOPMENT

- Flutter.
- FastAPI
- Flask
- Git
- Css
- Django
- React

DATA ANALYSIS

- Tableau
- PowerBI
- Power Automate

NETWORKING

- Cisco Packet Tracer.
- Nmap
- Metasploit
- Burpsuite

TOOLS

- VS Code
- Jupyter Notebooks
- Github
- MySQL
- SSMS
- Apache Netbeans
- Eclipse
- Microchip Studio
- Cisco Packet Tracer
- Arduino IDLE
- Fritze
- RStudio
- CodeBlocks
- Cursor
- SSMS
- Figma
- Gitbash
- Thonny.
- Kali Linux.
- Ubuntu

PROJECTS AND EXPERIENCE

AI Vein Locator – Developer

- Designed and implemented a computer vision tool to assist clinicians in locating veins for medical procedures.
- Applied deep learning and image processing techniques to improve accuracy in healthcare delivery.
- Enhanced clinical efficiency by providing a reliable, AI-driven solution for patient care.

Local Document Analysis Tool (UCC) – Developer

- Built an AI-powered system for document classification and analysis tailored to the Uganda Communications Commission.
- Automated regulatory workflows, reducing manual effort and improving reproducibility.
- Delivered a scalable solution that improved efficiency in document management and compliance.

FetalCare – Developer

- Developed an AI-powered fetal health analysis device integrating deep learning models with sensor data.
- Created a real-time monitoring system to support maternal healthcare and improve patient outcomes.
- Demonstrated the application of AI in critical healthcare environments.

Tomato Disease Detection System – Developer

- Engineered a deep learning-based computer vision model to identify tomato crop diseases.
- Empowered farmers with early detection tools, reducing losses and promoting sustainable agriculture.
- Applied TensorFlow and PyTorch frameworks to deliver accurate disease classification.

Smart Home Automation Solutions – Developer

- Designed IoT-based automation systems using Arduino, Raspberry Pi, and sensor integration.
- Built hardware-software interfaces and web-based controls for smart environments.
- Improved energy efficiency and user convenience through intelligent automation.

Other Projects include:

1. **Secure File Vault & Encrypted Chatroom – Developer**
2. **Network Scanner & Ethical Cybersecurity Lab – Developer**
3. **Local Document Summarizer – Developer**
4. **Bluetooth & QR Code File Transfer Tool – Developer**
5. **AI-Powered Network Reporting Tool – Developer and many others.**

LANGUAGES

- Luganda
- English
- Arabic

HOBBIES

- Chess
- Football
- Checkers
- Pool
- Researching
- Reading books

CAREER OBJECTIVE

To pursue a career as an **AI/ML Engineer** where I can apply my expertise in artificial intelligence, machine learning, IoT, and networking to design and implement intelligent, secure, and scalable systems. I aim to contribute to innovative organizations by leveraging my skills in Python, deep learning frameworks, embedded systems, and cybersecurity to build impactful solutions that address real-world challenges in healthcare, agriculture, smart environments, and digital ecosystems. My goal is to grow as a developer, researcher, and problem-solver while advancing technologies that empower users and promote accessibility.

REFEREES

1. **ENG. UMESH KUMAR**

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LUGOGO KAMPALA

2. **ENG. EDISON ATUHEIRWE**

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KISASI

DECLARATION

I **Tekirya Brighton** declare that the information given above are true reflection of my academic background, qualifications, skills and abilities.